

Hossein Moosavi Shoa'r

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PROFILE

Mechanical Engineer and M.Sc. student in Energy Conversion with research experience in thermal systems, heat transfer, and energy engineering. Skilled in MATLAB, Python, Fluent, numerical modeling, optimization, and experimental setup development. Author of journal and conference papers, seeking a PhD opportunity in energy storage and sustainable thermal systems.

EXPERIENCE

Teaching Assistant, design and optimization of thermal systems **spring 2026**

Sharif university of technology, department of mechanical engineering, Dr. H. Bahman Jahromi

Teaching Assistant, thermodynamics 2 **spring 2026**

Sharif university of technology, department of mechanical engineering, Dr. H. Pakzad

Teaching assistant, power plant design **fall 2025**

Sharif university of technology, department of mechanical engineering, Dr. H. Bahman Jahromi

Lab instructor, thermodynamics under-grads lab **fall 2025**

Sharif university of technology, department of mechanical engineering, Dr. M.H. Saidi

Teaching assistant, thermodynamics 1 **spring 2025**

Sharif university of technology, department of mechanical engineering, Dr. A. Moosavi

Lab instructor, fluid mechanics under-grads lab **spring 2025**

Sharif university of technology, department of mechanical engineering, Dr. B. farhanieh

Lab instructor, fluid mechanics under-grads lab **fall 2025**

Sharif university of technology, department of mechanical engineering, Dr. B. farhanieh

Teaching assistant, thermodynamics 1 **spring 2024**

Sharif university of technology, department of mechanical engineering, Dr. A. Moosavi

Teaching assistant, fluid mechanics 1 **fall 2023**

Sharif university of technology, department of mechanical engineering, Dr. A. Moosavi

Designing an application for designing vacuum system of street sweepers

fall 2023

Developed for TOMCOV company, supervised by Dr. M. R. Forouzan

skills: MATLAB programming, system modelling, fluid mechanics, turbomachinery

Designing air water generator using thermoelectric coolers, designing an application for easier future designs (bachelor project)

<https://doi.org/10.1016/j.ijft.2024.100740>

summer 2023

semi analytical modelling different type of heat exchangers integrated with TECs for analyzing efficiency of the related air water generator technology.

skills: MATLAB programming, heat and mass transfer analysis

Glider Competition (fly cup)

spring 2023

Placed 6th in the competition among 39 competitors, the glider was dynamically and aerodynamically analyzed and optimized and then built and tested.

skills: MATLAB programming, CFD simulation(fluent), CAD modeling (SolidWorks), aerodynamics, dynamics, mechanics of solids

Teacher assistant, Fluid mechanics 2

spring 2023

Isfahan university of technology, department of mechanical engineering, Dr. Saghafian

Apperentice at TOMCOV company

summer 2022

CAD modelling a new flow meter and a street sweeper

skills: CAD modelling (inventor)

Teacher assistant, fluid mechanics 1

spring 2022

Isfahan university of technology, department of mechanical engineering, Dr. MADADOLLAHI

Desingning car's exhaust heat recovery system (Rahneshan Tournament)

Finished 2nd.

fall 2021

skills: ansys fluent, heat transfer analysis, thermoelectric analysis, mechanics of solids, CAD modelling

Teacher assistant, mechanics of solids

summer 2021

Isfahan university of technology, department of mechanical engineering, Dr. Karevan

Teacher assistant, fluid mechanics 1

spring 2021

Isfahan university of technology, department of mechanical engineering, Dr.
MADADOLLAHI

EDUCATION

Sharif University of technology (GPA 19.02/20)

Tehran, Iran

Master of Energy Conversion (will be finished by 22 July of 2026)

2023-2026

Isfahan University of technology (GPA 18.99/20)

Isfahan, Iran

Bachelor of Mechanical Engineering

2019-2023

SKILLS

MATLAB PROGRAMMING (educated in MPT academy)

high intermediate

Python

low intermediate

C and C++

elementary

EES (engineering equation solver)

professional efficiency

SolidWorks – certified by Iran Technical & vocational training organization

high intermediate

Catia– certified by Iran Technical & vocational training organization

Inventor (learned in apprentice period)

low intermediate

AutoCAD

low intermediate

**Ansys – Space Claim, Ansys Meshing, ICM , Fluent, CFD post, Design Modeler,
UDF coding**

Intermediate

Star- CCM (taught by Dr. Pishevar in CFD course, bachelor)

Low intermediate

Aspen-HTRI (taught by Dr. Tavakoli-nezhad in heat exchanger design course)

Low intermediate

Microsoft Office

MISELANIOU RELEVANT SKILLS

English	high intermediate
Mathematical modelling	
Arduino Programming	
Welding and Visual Testing (VT)	
Turning machine operation	
Laser Cutting	
Teaching	
Teamwork	
Leading	
Searching	

HONOR AND AWARDS

Rank 3rd in 120 students of mechanical engineering, Isfahan University of Technology **summer 2023**

Direct Master to Sharif University of Technology through outstanding student selection procedure **summer 2023**

Chosen as top scientific student, Isfahan University of Technology **summer 2024**

PUBLICATIONS

Optimal Thermal Conditions for Longevity: A Thermodynamic Approach to Aging and Human Lifespan Enhancement

Still in progress

Uncertainty analysis of a high-temperature zeotropic cascade heat pump equipped with an ejector and phase separator (conference paper in ISME)

Recently accepted. Waiting for conference day.

Integral solution of the boundary layer of a power-law fluid over a cylinder considering variable properties and viscous dissipation (conference paper in ISME)

Recently accepted. Waiting for conference day.

Numerical investigation of enhanced cooling of a hot surface using an oscillating impinging jet and nanofluid

Recently accepted. Waiting for conference day.

Thermo-responsive nanostructured surface: Beeswax for enhanced condensation performance across solid, liquid, and transition states

<https://doi.org/10.1016/j.apsadv.2026.100936>

Presenting an analytical relationship for evaluating the performance of thermoelectric heat exchangers (conference paper in ISME)

<https://civilica.com/doc/2463143/>

Enhancement of dropwise condensation using biocompatible surfaces (conference paper in ISME)

<https://civilica.com/doc/2463143/>

Thermo-economic analysis of thermoelectric-based atmospheric water harvesters: Flow configuration impacts

<https://doi.org/10.1016/j.ijft.2024.100740>

VOLUNTARY RELEVANT WORKS

Member of Scientific Association of mechanical engineering, Isfahan University of Technology

Spring 2023

- Arranging festivals, tournaments, extra classes for students

Staff crew in International Conference on Energy Resilience and Sustainability (Tehran 2024)

Spring 2024